



Contact detection in computational mechanics: a signed distance field approach for convex superelliptical bodies

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Abstract

Accurate contact and impact detection remain key challenges in computational mechanics. This work presents a methodology for non-conformal contact detection between rigid convex bodies with superelliptical boundaries, represented using their respective Signed Distance Fields (SDFs). A method for approximating the SDF of discretized superelliptical boundaries is introduced, along with the concept of the Gap Distance Field (GDF). Reformulating the contact detection problem as an unconstrained minimization over the GDF simplifies the computation of minimum distance and overlap, while the closest contact point is obtained by projecting the SDF gradients. Numerical experiments involving superelliptical contact bodies validate the accuracy and robustness of the proposed approach. Additionally, the common normal (CN) condition naturally arises within this framework, and together with the optimality of the GDF minima, provides a sufficient condition for determining the minimum distance between the interacting bodies.

Keywords Contact detection · Gap distance field · Signed distance field · Minimum distance · Superellipse

1 Introduction

Contact detection is a fundamental challenge in computational mechanics, with significant implications across various fields of engineering, including robotics, structural analysis, and manufacturing. Rigid contact analysis addresses fundamental aspects, namely: defining an accurate geometric representation of the contacting surfaces, calculating the minimum distance between potential contact points and evaluation of the contact kinematics incorporating

the material properties of the bodies and pseudo-penetration depth [1].

Superellipses, the 2D analogue of superellipsoids, offer a compact and flexible geometric representation that has attracted significant attention in engineering applications. Their ability to model a broad spectrum of shapes makes them a versatile tool for representing solid geometries, seamlessly bridging the gap between elliptical and rectangular boundaries. This geometric flexibility makes superelliptical formulations particularly suitable for a variety of mechanics-related applications. In computational biomechanics, they are used to model anatomical structures and simulate contact interactions [2–4]. In vehicle simulation, superellipses provide an effective means to approximate vehicle outlines and collision boundaries [5, 6]. They are also widely used in granular and particle dynamics to capture shape effects in packing and flow behavior [7–10]. In robotics, superelliptical shape representation has been applied to grasping and interaction modeling [11], as well as to quasi-static mechanical manipulation of rigid bodies [12]. Finally, they have also found application in beam-to-beam contact modeling, where superelliptical formulations have been used to describe beam cross sections [13, 14].

The primary goal of contact detection algorithms is to accurately determine whether two shapes are in contact or

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not and to compute the point of contact or, to compute the minimum distance between them [15]. Contact between two convex superellipses is non-conformal, hence, from the boundary representations of the colliding bodies, along with their positions and spatial orientations, the resolution of the minimum distance problem becomes a purely geometric task [16].

Although (apparently) simple in terms of geometric description, minimum distance determination between superellipses lacks closed-form solutions. Consequently, many techniques have been developed throughout the years but rely heavily on iterative and non-linear optimization solvers. Recently, Signed Distance Fields (SDF) have emerged as a powerful tool for representing complex geometries and their spatial properties. SDFs encode the shortest distance from any point in space to the surface of an object and can be effectively leveraged for contact detection between convex superelliptical bodies. This approach allows for the efficient and robust identification of contact points, minimum distance and penetration, crucial for accurately modeling the interactions between contacting bodies. However, the computation of the intersection between two arbitrary bodies defined by their SDFs remains largely unexplored topic [17]. Generating accurate SDFs for superelliptical boundaries presents its own set of challenges. Addressing these challenges is critical for ensuring precise and efficient contact detection.

This work introduces a methodology for non-conformal contact detection between rigid convex bodies described by superelliptical boundaries. The proposed approach employs SDF representations of the bodies and formulates the contact detection problem as an unconstrained minimization of a newly defined Gap Distance Field (GDF). This re-formulation simplifies the computation of minimum distance and pseudo-penetration depth. Additionally, the framework naturally incorporates the concept of common-normal with

strong geometrical constraints, ensuring that the identified contact points are indeed the closest pairs.

The paper is organized as follows. Section 2 provides a review of related works, Sect. 3 details the formulation of superellipses while Sect. 4 provides an introductory insight on SDF's. Section 5 presents the proposed methodology, including the approximate generation of SDFs and the definition of the Gap Distance Field. Section 6 presents numerical validation through various test cases involving superelliptical contact scenarios. Finally, Sect. 7 discusses the implications of the results and concludes with potential avenues for future research.

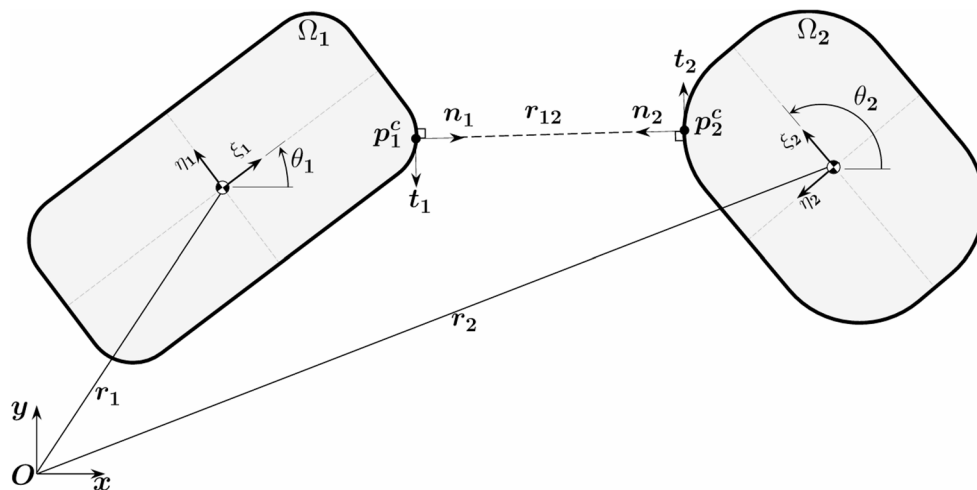
1.1 Problem statement

Given two convex bodies Ω_1 and Ω_2 , modeled as superellipses defined by their respective shape and spatial parameter the objective is to determine the gap (separation) or pseudo-penetration (overlap) between these bodies when they are separated or overlapping. The problem is schematically presented in Fig. 1.

In separation, the task involves identifying the closest contact points on the boundaries of the two bodies. These points correspond to the locations where the gap equals the minimum distance, which is crucial for accurately modeling physical interactions. When overlapping occurs, these points help quantify the interpenetration, which must be corrected during the contact resolution stage [18]. The general problem of finding the contact pair p_1^c, p_2^c on boundary $\partial\Omega_1$ and $\partial\Omega_2$, respectively, for which the distance is minimized, can be expressed as a constrained optimization as:

$$p_1^c, p_2^c = \arg \min_{\substack{p_1 \in \partial\Omega_1 \\ p_2 \in \partial\Omega_2}} \frac{1}{2} \|r_{12}(p_1, p_2)\|^2 \quad (1)$$

Fig. 1 Illustration of a contact detection problem for two arbitrary convex bodies defined as superellipses



where $\|(\cdot)\| = \sqrt{(\cdot)^T(\cdot)}$ is the Euclidean norm of $(\cdot)$. Here, r_{12} is a vector connecting the two points located on the boundary of each superellipse imposed by the two boundary equality constraints. Although conceptually simple, the problem exhibits inherent nonlinearity, rendering linear techniques ineffective. Additionally, the problem is highly structured, and general nonlinear optimization methods that do not leverage this structure may likewise prove inefficient [19].

1.2 Contributions

This work presents an alternative approach for non-conformal contact detection between rigid convex bodies with superelliptical boundaries. The main contributions of the present work are: (1) the adoption of a SDF-to-SDF contact detection strategy, and (2) the formulation of the contact detection problem as an unconstrained minimization of a newly defined Gap Distance Field (GDF) eliminating the need of additional geometrical constraints as the framework naturally integrates the concepts of minimal distance.

2 Related word

Contact detection for convex bodies, specially for the ellipsoidal family of shapes, has been widely studied, leading to the development of numerous approaches to address the challenges of accurately and efficiently determining contact conditions.

The primary objectives of contact detection algorithms include determining the contact status between two bodies, computing the minimum separation distance or maximum penetration depth overlapping, and computing contact parameters such as closest points/contact pair and contact normal vectors. These algorithms generally fall into two categories: intersection/collision detection and proximity queries.

Early research in collision detection focused on simpler geometric shapes, such as spheres, axis-aligned bounding boxes (AABBs) [20], or oriented bounding boxes (OBBs) [21], due to their computational efficiency. However, the need for more realistic and flexible representations led to the exploration of ellipsoids and superellipsoids. While the geometrical versatility of superellipsoids made them attractive for applications in computer graphics and physical simulations, they also introduced additional challenges in collision detection.

Initial approaches derived algebraic methods to determine exact intersection conditions by solving the implicit equations of the involved bodies often relying on

solving algebraic systems, including characteristic polynomial equations.

Researchers such as Eberly [22] proposed a method based on computing roots to quartic polynomials, while Alberich-Carramiñana et al. [23] demonstrated that the problem reduces to studying the nature of the roots of a cubic polynomial to classify the relative position of two ellipses. Brozos-Vázquez et al. [24] extended this concept to contact between small ellipsoids and other quadrics, while Wang et al [25] and Jia et al. [26] showed that contact status can be determined algebraically by examining the eigenvalues of the characteristic polynomial of a given pair of ellipsoids. Despite their theoretical rigor and effectiveness, algebraic methods often struggle to compute separation distances and generalize poorly to non-elliptical geometries.

To address limitations of computational instabilities due to solving the ill-conditioned equations to find the intersections of two ellipses, Lin and Ng [27, 28] proposed a geometric potential-based contact method by solving two decoupled minimization problems to identify the point in each ellipse that minimizes the potential field of the other. Kheradmand et al. [29] presented a similar approach that falls within the class of geometric potential methods but proposed a geometrical equivalency operation that transforms an ellipsoid-ellipsoid contact to an equivalent ellipsoid-sphere contact problem.

Another class of contact detection methods is based on the concept of the CN. First introduced also by Ng [28], the CN condition states that, at the closest points between the two bodies, the outward-pointing normal vectors at these points must align along the same straight line. This property naturally characterizes the minimum distance condition. Wellmann et al. [30] developed a 2D unconstrained optimization formulation based on this concept by formulating the problem of contact detection in terms of the contact direction angles.

Portal et al. [31] developed a constrained optimization formulation involving surface normal alignment at contact points, while Rebel and Fröhlich [32] employed an iterative Levenberg-Marquardt-based procedure for parametric ellipsoids, using the normal alignment conditions as convergence criteria. On the other hand, Lopes et al. [16] tackled a similar problem for implicit superellipsoids but employed algebraic and differential geometry to derive conditions for minimum distance from the anti-parallelism of the normal vectors. This set of non-linear equations was numerically solved using the Newton–Raphson method with analytical Jacobian matrices.

Iterative approaches have also been explored, such as the method by Li et al. [33] which translated the contact detection problem into the solution of finding a scaling factor for tangency of the superellipses. Recent advancements

in contact detection for arbitrarily shaped bodies include machine learning-based methods using classification and regression neural networks [34–36].

The use of SDFs for collision detection, though not new, has gained traction in computer graphics, robotics, and computational mechanics due to their superior query efficiency and ability to represent continuous geometries. Macklin et al. [37] introduced a local optimization algorithm to determine the closest distance between a SDF and mesh elements and López-Adeva Fernández-Layos and Merchante [18] presented an algorithm, based on the ellipsoid method, to compute the minimum distance and penetration depth between two convex bodies represented by SDFs. Lai et al. [38] developed an approach for the discrete element method (DEM) modeling of generic irregular-shaped particles built on a node-to-SDF collision detection. Areias et al. [39] presented an alternative to classical geometrical formulations by introducing approximate DFs computed through a PDE approach using the screened Poisson equation. This PDE-based distance computation method was originally introduced by Crane et al. [40, 41] for geodesic distance computation.

For an extensive overview of contact detection methods applied to convex bodies, comprehensive reviews in [42–44] provide valuable insights into both classical techniques and emerging approaches.

3 Superellipse formulation

Originally introduced by French mathematician Gabriel Lamé [45] the superellipse, or Lamé curve, represents a generalization of the ellipse equation, preserving the geometric features of semi-major axis, a semi-minor axis, and symmetry about these axes. The implicit representation of the superellipse, formulated in the body's fixed frame of reference (FoR), is given by:

$$f_{SE}(x, y, \lambda) = \left| \frac{x}{a} \right|^{\frac{2}{\varepsilon}} + \left| \frac{y}{b} \right|^{\frac{2}{\varepsilon}} = C, \quad (2)$$

$$\lambda = \{a, b, \varepsilon\} \in \mathbb{R}^+$$

The parameters $\lambda = \{a, b, \varepsilon\}$ are the shape parameters and, for the present case, are always considered as positive real numbers. The parameters a and b are the semi-axes lengths of the superellipse in the direction of the principal axes and, without loss of generality, we define $a \geq b$ within the scope of this study. The parameter ε defines the “squareness” of the resulting shape. Variable C corresponds to the equipotential surface value of the superellipse, which will henceforth be considered equal to 1. This formulation describes a closed boundary inscribed in a rectangle defined by corners

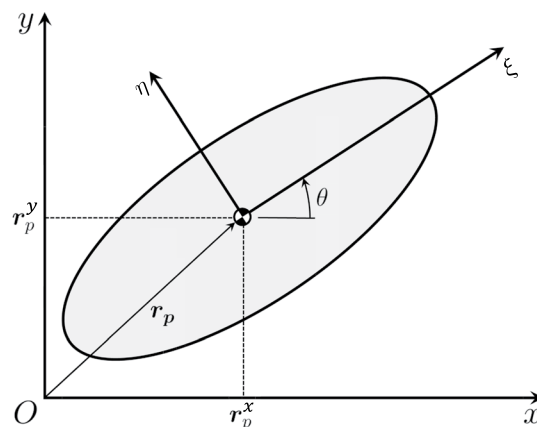


Fig. 2 Superellipse ($\varepsilon = 1$) in global coordinate system (O, x, y) with local coordinate system (O, ξ, η) centered at r_p and rotated by θ in the x -axis about its centroid

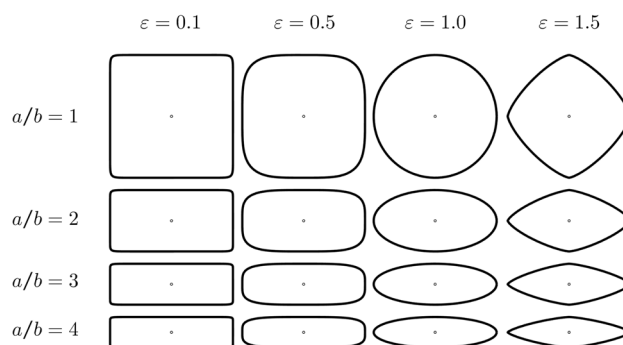


Fig. 3 Superellipse shape variation controlled by shape parameters $\lambda = \{a, b, \varepsilon\}$

$[-a, -b]$ and $[a, b]$. For non-centered and non-axis aligned superellipses, the FoR of each body can be described via the reference point coordinates (origin of body FoR) [46]. The vector of reference point coordinates $q = \{r, \theta\}$ consists of the position vector r and the rotation angle θ of the body FoR w.r.t the global FoR, as illustrated in Fig. 2.

Superellipses (and their higher dimensional generalization) give an excellent trade-off between model complexity and shape flexibility by simply changing 3 parameters in Eq.(2), as shown in Fig. 3.

The use of “squareness” parameters $0 < \varepsilon < 2$ ensures strict convexity of the resulting boundary, allowing for a smooth transition from rectangular shapes ($\varepsilon \rightarrow 0$) to elliptical ones ($\varepsilon = 1$). This approach also avoids singularities in the definition of the normal direction, which would otherwise arise in the presence of non-smooth boundaries [13]. To prevent difficulties due to these singularities, this parameter can be further bounded as $0.1 < \varepsilon < 1.9$ [47]. Note that a standard ellipse can be obtained by setting $\varepsilon = 1$ while a perfect circle can be obtained by additionally imposing $a = b$. Additionally, their ability to represent a wide variety

of shapes encountered in real-world scenarios and mechanical systems, superellipses are widely used in the context of computational mechanics and computer graphics.

Alternatively, the implicit formulation, Eq.(2) can be rewritten explicitly as the following parametric expression for a standard superellipse:

$$s_{SE}(\varphi, \lambda) = \begin{cases} x = a \cdot \cos^\varepsilon(\varphi) \\ y = b \cdot \sin^\varepsilon(\varphi) \end{cases} \quad \varphi \in [0, 2\pi[\quad (3)$$

where $s_{SE} : \mathbb{R}^1 \rightarrow \mathbb{R}^2$ is the mapping function from the parameter domain to the physical domain, the exponentiation operator ε corresponds to a signed power operator such that $(\cdot)^\varepsilon = \text{sgn}(\cdot) \cdot |\cdot|^\varepsilon$ for $(\cdot) \in \{\cos \varphi, \sin \varphi\}$; and φ corresponds to the parametric argument (with no geometric interpretation).

Superellipses have several important points located at known coordinates such as extremum points, located at $(\pm a, 0)$ and $(0, \pm b)$; and its four diagonal “corner” points $(\pm \sigma a, \pm \sigma b)$, where $\sigma = 2^{-\varepsilon/2}$ is called the “supereness” parameters [48].

The superellipse can be extended to higher dimensions while preserving its fundamental mathematical structure. Specifically, the superellipsoid, as introduced by Barr [49] represents the three-dimensional generalization of the superellipse. However, in the context of this work, we focus exclusively on superellipses, with the exploration of superellipsoids left for future research.

By applying a rotation θ to the standard superellipse formulation Eq.(3) and translating it by its non-zero centroid vector $\mathbf{r}_p$, the new parametric formulation can be compactly expressed, in homogeneous coordinates, as:

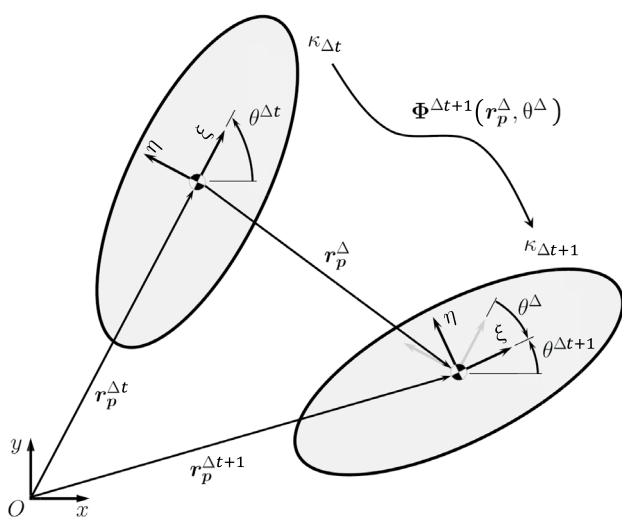


Fig. 4 Rigid body transformation of a superellipse in current configuration $\kappa_{\Delta t}$ to next configuration $\kappa_{\Delta t+1}$ through spatial transformation $\Phi^{\Delta t+1}(\mathbf{r}_p^\Delta, \theta^\Delta)$

$$\mathbf{x} = \Phi \cdot \mathbf{S} \cdot \boldsymbol{\varepsilon}_\varphi \quad (4)$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad \Phi = \begin{bmatrix} \cos \theta & -\sin \theta & r_p^x \\ \sin \theta & \cos \theta & r_p^y \\ 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{S} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\varepsilon}_\varphi = \begin{bmatrix} \cos^\varepsilon(\varphi) \\ \sin^\varepsilon(\varphi) \\ 1 \end{bmatrix}$$

The matrix Φ is called the homogeneous transformation matrix which encompasses both rotational and translational transformations; the matrix $\mathbf{S}$ represents the diagonal matrix of semi-axis lengths; and $\boldsymbol{\varepsilon}_\varphi$ is the vector of unit parametric coordinates.

Within the scope of the present study, superelliptical bodies are treated as rigid bodies. The kinematics of the superellipse is then fully described by the translation of its center of mass and the rotation about its centroid. For simplicity, we restrict the analysis to cases where only prescribed motion and rotation are applied, without considering external forces or torques apart from gravitational force.

We define the current configurations κ with superscript Δt , while the next configurations are denoted by $\Delta t + 1$. Figure 4 depicts the relevant quantities for the presented rigid body transformation. The instantaneous configuration of an arbitrary superelliptical body is given by the transformation Φ of the reference superellipse. The next configuration $\kappa_{\Delta t+1}$ is then defined by a new centroid vector $\mathbf{r}_p^{\Delta t+1}$ and new orientation $\theta^{\Delta t+1}$. These quantities can be updated from the current configuration $\kappa_{\Delta t}$ at time $\Delta t + 1$ using first-order Euler integration. The updated configuration $\kappa_{\Delta t+1}$ is determined by the new homogeneous transformation matrix $\Phi^{\Delta t+1}$. The new rotation and translation components of $\Phi^{\Delta t+1}$ are given by the update of the current configuration as follows:

$$\mathbf{r}_p^{\Delta t+1} = \mathbf{r}_p^{\Delta t} + \mathbf{r}_p^\Delta \quad (5)$$

$$\theta^{\Delta t+1} = \theta^{\Delta t} + \theta^\Delta \quad (6)$$

The incremental translation $\mathbf{r}_p^\Delta$ and rotation θ^Δ represent the changes in position and orientation from Δt to $\Delta t + 1$ and are computed as:

$$\mathbf{r}_p^\Delta = \dot{\mathbf{r}}_p^\Delta \Delta t \quad \text{with} \quad \dot{\mathbf{r}}_p^\Delta = \dot{\mathbf{r}}_p^{\Delta-1} + \mathbf{g} \Delta t \quad (7)$$

$$\theta^\Delta = \dot{\omega}^0 \Delta t \quad (8)$$

4 Signed distance field

Given a closed body $\Omega \subset \mathbb{R}^n$, with boundary $\partial\Omega$, the Euclidean distance between an arbitrary query point in space $\mathbf{x}_q \in \mathbb{R}^n$ and $\partial\Omega$ is given by:

$$d(\mathbf{x}_q, \partial\Omega) = \inf_{\mathbf{y} \in \partial\Omega} \|\mathbf{x}_q - \mathbf{y}\| \quad (9)$$

The Jordan curve theorem states that, as long as given a boundary in $\mathbb{R}^2$ forms a closed loop with no self-intersections, it partitions the entire $\mathbb{R}^2$ space into an "interior" region bounded by the boundary and an "exterior" region containing both nearby and distant exterior points [50]. Consequently, the Unsigned Distance (UD), resulting from Eq.(9), can be modified by introducing a sign operator that encompasses the notion of interior/exterior of a given queried point to a closed boundary. This gives rise to the notion of Signed Distance, hereinafter simply referred to as SDF, defined as a scalar function $\phi(\mathbf{x}_q) : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $\forall \mathbf{x}_q \in \mathbb{R}^n$:

$$\phi(\mathbf{x}_q, \partial\Omega) = \text{sgn}(\mathbf{x}_q, \Omega) \cdot d(\mathbf{x}_q, \partial\Omega) \quad (10)$$

The *signum* function $\text{sgn}(\mathbf{x}_q, \Omega)$ is the boundary's inside-outside query function and is defined as:

$$\text{sgn}(\mathbf{x}_q, \Omega) := \begin{cases} -1, & \text{if } \mathbf{x}_q \in \Omega \\ 0 & \text{if } \mathbf{x}_q \in \partial\Omega \\ +1, & \text{otherwise} \end{cases} \quad (11)$$

The SDF then gives an implicit representation of the domain of the geometry since its boundary is the zero-level set of the SDF, denoted as $\phi(\mathbf{x}) = 0$ while the field values $\phi(\mathbf{x})$ describe the signed distance to the boundary with $\phi(\mathbf{x}) < 0$ representing the interior of the domain. The isolines of the SDF represent equidistant level sets of $\phi(\mathbf{x})$ with constant

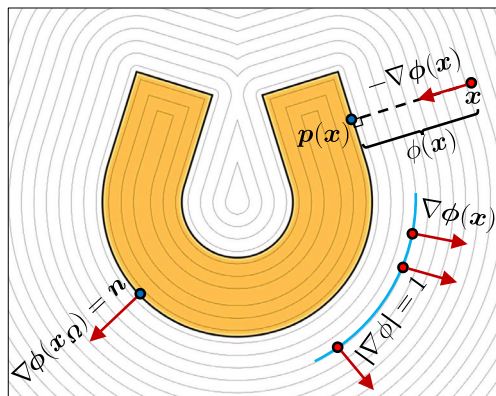


Fig. 5 Example of a SDF and its geometrical properties for a generic boundary (adapted from [37])

distance value. The introduction of the sign to the distance field also removes the C^1 -discontinuity present in the unsigned distance field at the boundary. As a result, the gradient vector field exhibits a continuous and smooth flow across the boundary of the domain.

A fundamental property of the SDF is that its gradient $\nabla\phi(\mathbf{x})$ is perpendicular to the level sets of $\phi(\mathbf{x})$ and points outward for regions outside the boundary $\phi(\mathbf{x}) > 0$ and inward for regions inside $\phi(\mathbf{x}) < 0$. Another critical property of the SDF is that its gradient magnitude satisfies the Eikonal equation $|\nabla\phi(\mathbf{x})| = 1$ almost everywhere, ensuring that the distance is represented consistently across the field. Additionally, for any point $\mathbf{x}$ in the domain, the closest point on the boundary, denoted as $\mathbf{p}(\mathbf{x})$, is readily obtained through gradient projection. At the boundary, where $\phi(\mathbf{x}) = 0$, the gradient $\nabla\phi(\mathbf{x})$ corresponds directly to its outward normal vector $\mathbf{n}(\mathbf{x})$. These characteristics are depicted in Fig. 5, demonstrated in the context of a generic boundary and its associated SDF. Furthermore, for convex boundaries, additional properties arise, such as: $\mathbf{p}(\mathbf{x})$ is unique, $\forall \mathbf{x}: \phi(\mathbf{x}) > 0$; and if Ω is convex then $\phi(\mathbf{x})$ is also convex [18]. These last specific properties for convex boundaries will prove to be particularly important for proposed contact detection methodology.

Most common algorithms for calculating the SDF involve the numerical resolution of the Eikonal equation. These include the fast marching method [51] and fast sweeping method [52].

5 Proposed methodology

5.1 Overview

The proposed methodology involves three main steps that will be further discussed: generation of the approximate SDF for superellipses; the generation of the Gap Distance Field and the computation contact pair/closest distance points.

5.2 Approximate SDF

While analytical forms of SDFs are readily available for simple shapes, such as circles or squares, extending these formulations to more complex shapes can be significantly more challenging.

Although the superellipse boundary can be defined implicitly through its mathematical equation, directly computing the SDF from this implicit form poses significant challenges, particularly for applications requiring efficient distance evaluations. The implicit equation of a superellipse provides an inside-outside indication for a given point or

a given level-set value, but does not inherently yield the shortest distance to the boundary. While the exact Euclidean distance $\phi(\mathbf{x}_q, \partial\Omega)$ between a query point and a superellipse can be computed through numerical optimization methods, which is computationally expensive and impractical for large-scale simulations or real-time applications, no closed-form algebraic expression for this distance remains is known.

Some authors use an approximation, called radial Euclidean distance or Algebraic distance, whose explicit expression can be derived based on the implicit equation of the superellipse [53, 54]. This distance is defined as the distance between the query-point $\mathbf{x}_q$ and the intersection point of the line passing through $\mathbf{x}_q$ to the centroid of the superellipse, with its boundary, but fails to return minimum distance from the query point.

Unlike implicit representations, parametric surfaces allow for a more straightforward generation of geometric representations such as polygonal meshes. This property makes them a preferred choice in computational geometry and various design applications [55]. Through discretization, one can assume that an arbitrary convex boundary $\partial\Omega$ can be approximated by a n -sided convex polytope as the union of n piecewise linear boundary elements $e_i = [\mathbf{x}_i, \mathbf{x}_{i+1}]$, $i = 1, \dots, n$, with $n + 1$ being the total number of vertices/nodes of the discretized curve [56], as:

$$\partial\Omega \approx \partial\tilde{\Omega} := \bigcup_{i=1}^n e_i \quad (12)$$

Furthermore, this approach can highly benefit from techniques such as adaptive refinement, where local resolution can be determined by the proximity to boundaries. Additionally, discretization of the superellipse boundary through its parametric formulation in Eq.(3), employing uniform sampling of the parameter φ inherently adapts the local element density based on the curvature of the boundary [57]. This approach naturally results in a higher concentration of elements in regions with increased curvature, where greater

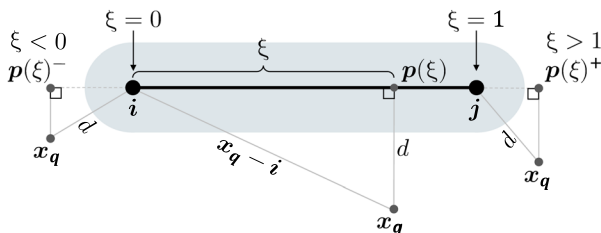


Fig. 6 Schematic representation for the computation of a simplicial linear element SDF: $\mathbf{p}(\xi)$ point represents the CPP within the parametric domain of the element while $\mathbf{p}(\xi)^+$ and $\mathbf{p}(\xi)^-$ represent the projection points outside the parametric domain of the linear element

resolution is typically required to accurately capture geometric features without additional computational procedures for local adaptive refinement. The distance field of an element e_i quantifies the shortest distance from a query point $\mathbf{x}_q \in \mathbb{R}^2$ to a piecewise linear element defined by its nodes i and j . The parametrization of a given point within the domain of e_i can be defined as $\mathbf{p}(\xi) = \mathbf{i} + \xi(\mathbf{j} - \mathbf{i})$, with $\xi \in [0, 1]$. The minimum distance of $\mathbf{x}_q$ to e_i is given by its closest point projection (CPP) through orthogonal projection onto the element e_i . The parametric coordinate ξ of the CPP is given by:

$$\xi = \frac{\langle \mathbf{x}_q - \mathbf{i}, \mathbf{j} - \mathbf{i} \rangle}{\|\mathbf{j} - \mathbf{i}\|^2} \quad (13)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product of two vectors and $\|\mathbf{j} - \mathbf{i}\|^2$ is the squared length of the element e_i . The distance between the query point and the correspondent CPP can then be computed as follows:

$$d(\mathbf{x}_q, e_i) = \|\mathbf{x}_q - (\mathbf{i} + (\xi \cdot \mathbf{j} - \mathbf{i}))\| \quad (14)$$

The above equation describes a hyperplane-based approach to the unsigned distance field (UDF) of a linear element, dividing the domain into two distinct halfspaces, where the distance grows orthogonally relative to the defined hyperplane. The resulting UDF is depicted in Fig. 7(a). An alternative approach includes defining an embedding hyperplane to e_i , characterized by its outward normal and offset, and directly computing the distance by the norm of the projection vector, see [18]. However, the distance field resulting from Eq.(14) is only represented correctly within the range of parameter ξ . For points outside this range ($\xi < 0$ & $\xi > 1$), the CPP of $\mathbf{x}_q$ should correspond to the closest node of e_i . Figure 6 depicts schematically the parameters involved in the definition of the true distance field of e_i .

Hence, for a finite linear element, the parametric coordinate of the CPP for both inside and outside of the parametric limits can be unified by clamping the parameter ξ to the $[0, 1]$ interval, which confines the projected point to the boundaries of the element, yielding to:

$$\xi' = \min \left(1, \max \left(0, \frac{\langle \mathbf{x}_q - \mathbf{i}, \mathbf{j} - \mathbf{i} \rangle}{\|\mathbf{j} - \mathbf{i}\|^2} \right) \right) \quad (15)$$

The UDF is then computed by substituting ξ' into Eq.(14). The resulting distance field is depicted in Fig. 7(b). Sukumar and Srivastava [58] presented an alternative way to approximate the UDF of a linear element by interpreting it as the trimming of an infinite line by a disk of radius of half element length; however, this approximation fails to preserve distance regularity of the isolines.

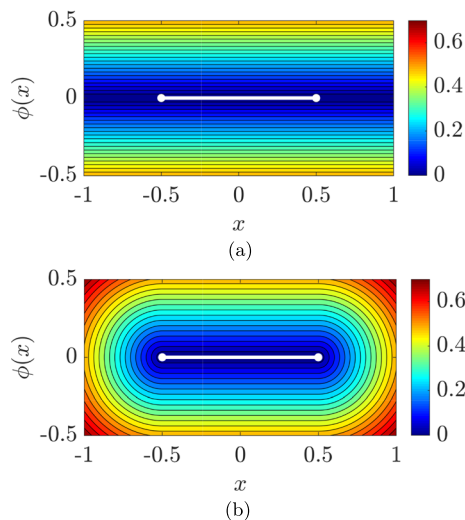


Fig. 7 Distance field for a linear element defined by **a** halfspace approach with ξ given by Eq.(13) and **b** exact distance with ξ given by Eq.(15)

To incorporate the sign into the computed UDF, a geometric convention is used to define whether $\mathbf{x}_q$ lies above or below the oriented element. This can be achieved either by computing the sign of the dot product of $\langle \mathbf{x}_q - \mathbf{i}, \mathbf{j} - \mathbf{i} \rangle$ or, equivalently, by computing the signed area of the parallelogram formed by these vectors [59].

Constructive Solid Geometry (CSG) allows to achieve complex shape using a collection of simplicial shapes which are combined via Boolean operations such as intersection, union, or difference [60]. Analogously to what is defined in Eq.(12), an approximation of the original convex body's SDF, can be obtained as the combination of the distance fields of each element of its discretized boundary:

$$\phi(\mathbf{x}_q, \partial\Omega) \approx \tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega}), \quad \partial\tilde{\Omega} = e_1, \dots, e_n \quad (16)$$

This discretized approach offers several advantages. First, it reduces the complexity of distance computation by enabling the use of closed-form expressions for the SDF of each boundary element, avoiding the need for iterative or numerical optimization to find the closest boundary point to the implicit form of the bodies. Second, discretization provides a straightforward way to approximate the global SDF of the superellipse by aggregating the contributions of individual element SDFs, which can be efficiently computed and combined. The approximate SDF accuracy is, however, highly dependent on the discretization level employed. Depending on whether a UDF or SDF is computed for each boundary element, the resulting SDF for the approximate boundary can be determined by the combination of the single boundary elements' SDFs via the following Boolean operations, respectively:

$$\begin{aligned} \tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega}) &= \text{sgn}(\mathbf{x}_q, \partial\tilde{\Omega}) \cdot \min_{i=1, \dots, n} d(\mathbf{x}_q, e_i) \text{ or} \\ \tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega}) &= \max_{i=1, \dots, n} \phi(\mathbf{x}_q, e_i) \end{aligned} \quad (17)$$

The implicit boundary representation of Eq.(2) can be confidently used as the *signum* function for the discretized superellipse provided that a fine discretization of the original implicit boundary is used. The implicit function of Eq.(17) satisfies the exact distance condition at discretization nodes $\tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega}) = \phi(\mathbf{x}_q, \partial\Omega)$, $\forall \mathbf{x} \in \partial\tilde{\Omega} \cap \partial\Omega$ and bounds the exact solution on nodes other than boundary's $\tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega}) > \phi(\mathbf{x}_q, \partial\Omega)$, $\forall \mathbf{x} \notin \partial\tilde{\Omega} \cap \partial\Omega$. This suggests that the piecewise linear boundary discretization leads to an upper bound approximation of the exact SDF. Precision and correctness are guaranteed at nodal points which can be useful for node-based contact detection strategies such as in [61, 62]. The algorithm for computation of the approximate SDF of a query point to a discretized boundary of a superellipse is explained in the following pseudo-code:

```

1: Inputs:  $a, b, \varepsilon, \mathbf{r}, \theta, n, \mathbf{x}_q$ 
2: Initialize UDF:  $d(\mathbf{x}_q, \partial\tilde{\Omega}) = \infty$ 
3:  $\partial\Omega \approx \partial\tilde{\Omega}(n) \triangleright \text{Eq. (3)}$ 
4: for  $k = 1 : n$  do
5:   Get nodes  $(\mathbf{i}, \mathbf{j})_{e_k}$ 
6:    $d(\mathbf{x}_q, e_k) \triangleright \text{Eq. (14) and Eq. (15)}$ 
7:    $d(\mathbf{x}_q, \partial\tilde{\Omega}) \leftarrow \min(d(\mathbf{x}_q, e_k), d(\mathbf{x}_q, \partial\tilde{\Omega}))$ 
8: end for
9:  $\tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega}) \leftarrow \text{sgn}(\mathbf{x}_q, \partial\Omega) \cdot \tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega})$ 
10: return  $\tilde{\phi}(\mathbf{x}_q, \partial\tilde{\Omega})$ 

```

Fig. 8 Approximate Signed Distance of Query Point to Superellipse

As motivation, the exact SDF of a unit length square, represented in Fig. 9(a), is compared to the approximated SDF for hyperplane SDF, Fig. 9(b) and exact SDF, Fig. 9(c) of the simplicial linear element. The exact SDF formula of an arbitrary centered and axis-aligned rectangle is well known and given by $\phi = \min(\max(\Delta x, \Delta y), 0) + \|\max(\Delta x, 0), \max(\Delta y, 0)\|$, where the operator Δ denotes the directional difference between the coordinate of the query point and the semi-axis of the rectangle. The present example clearly shows the correctness of the piecewise linear approximation of the rectangle when the clamped parametric projection parameter is used to obtain the individual SDFs before union. On the other hand, the halfspace approach fails to correctly represent the outside SDF in corner regions, although the interior of the rectangle is correct. The proposed SDF approximation methodology was then evaluated for two superellipses defined by their respective shape and spatial parameters, as illustrated in Fig. 10. The first superellipse, Fig. 10(a) is characterized by $\lambda_1=1, 1, 0.2$ and $\mathbf{q}_i = \{0, 0, 0\}$

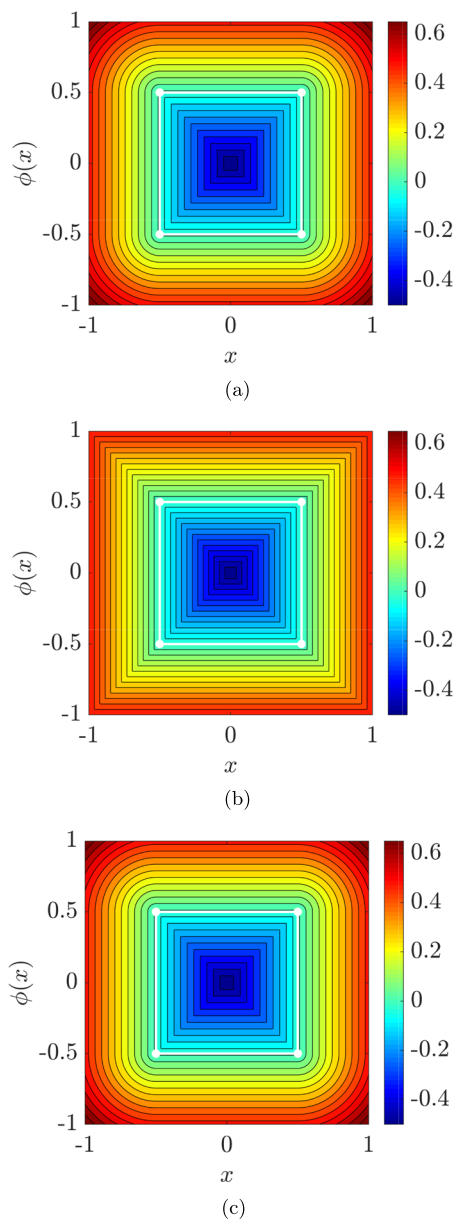


Fig. 9 **a** Exact SDF of unit square, **b** Resultant SDF obtained through combination of halfspace SDFs of Eq.(13), and **c** Resultant SDF obtained through exact linear element SDF combination of Eq.(15)

while the other, Fig. 10(b), is defined by $\lambda_j = \{1, 1.5, 1\}$ and $q_j = \{2, 1.7, 0.3\}$. A discretization level of $n = 1000$ is employed to sample their boundaries uniformly in the parametric space. SDF results presented in Fig. 10 showed excellent agreement with the expected distance field, validating the method's ability to handle superelliptical geometries.

5.3 Gap distance field and apparent contact point

As discussed in Sect. 2, a recurrent approach for solving the minimum distance problem relies on the CN condition.

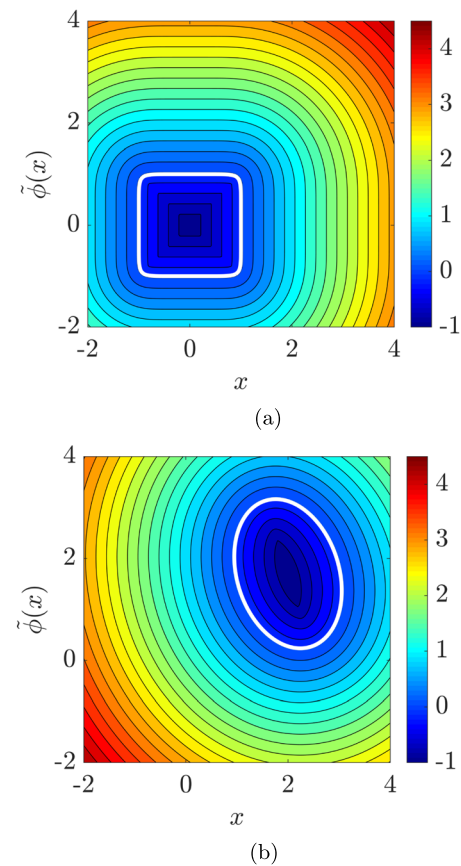


Fig. 10 Approximate SDF for superellipse characterized by **a** $\lambda_i = \{1, 1, 0.2\}$ and $q_i = \{0, 0, 0\}$ and **b** $\lambda_j = \{1, 1.5, 1\}$ and $q_j = \{2, 1.7, 0.3\}$

However, this condition alone only provides a weak constraint of the distance problem, representing a necessary but not sufficient condition of minimum distance as multiple contact pairs might satisfy the CN condition without guarantee of global minima [63]. In fact, for non-conformal contact scenarios, there can be, at most, four possible contact pair solutions of the CN condition problem [55]. To bridge this limitation and ensure that the computed distance is indeed minimal, the problem can be augmented with additional geometric constraints. By transforming the primal formulation of Eq. (1) into its Lagrangian equivalent results in:

$$\mathcal{L}(\mathbf{p}_1, \mathbf{p}_2, \nu_1, \nu_2) = \frac{1}{2} \|\mathbf{r}_{12}\|^2 + \nu_1 f_1(\mathbf{p}_1) + \nu_2 f_2(\mathbf{p}_2) \quad (18)$$

where ν_1, ν_2 are the Lagrange multipliers, and f_1, f_2 are the boundary equality constraints that can be defined by some implicit boundary representation. The stationarity condition of the Lagrangian function in Eq.(18) yields the following

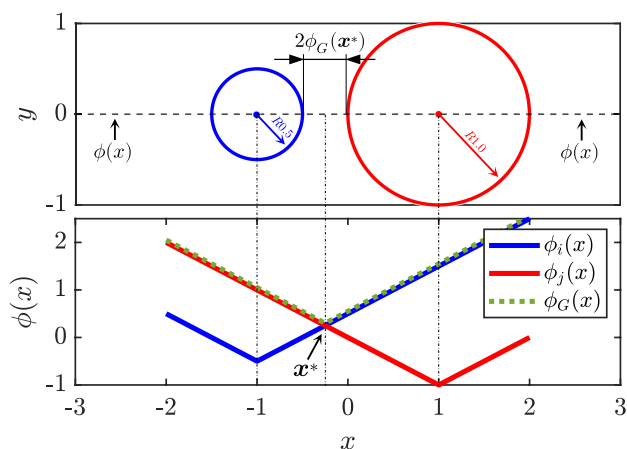


Fig. 11 Illustration of the minimum distance problem as a minimization of the Gap Field ϕ_G over a 1D cut. The example shows the squared SDFs ($\phi_{i,j}$) (solid lines) of two circular boundaries with their resulting Gap Field (dashed line). The minimum of ϕ_G corresponds to half the minimum distance between source points

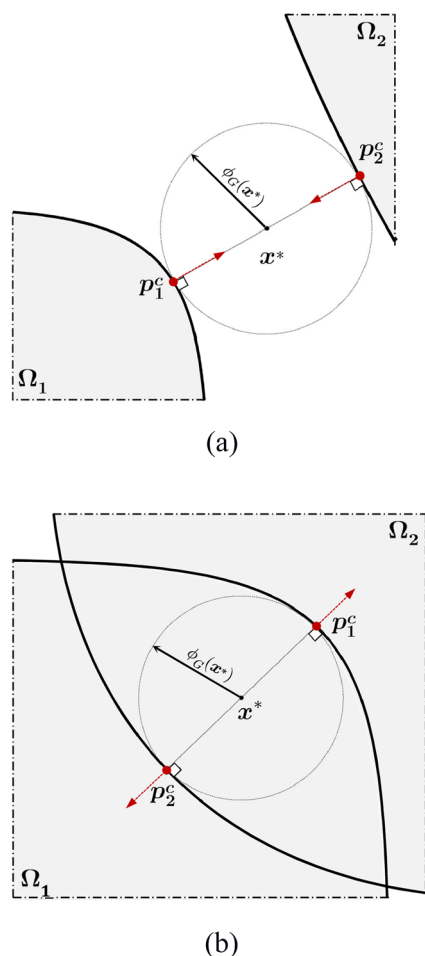


Fig. 12 **a** Separation scenario where minimum distance is given by $2\phi_G(x^*)$ and **b** overlap scenario with representation of the maximum inscribed circle centered at the CC x^* . Overlap distance is then given by $-2\phi_G(x^*)$

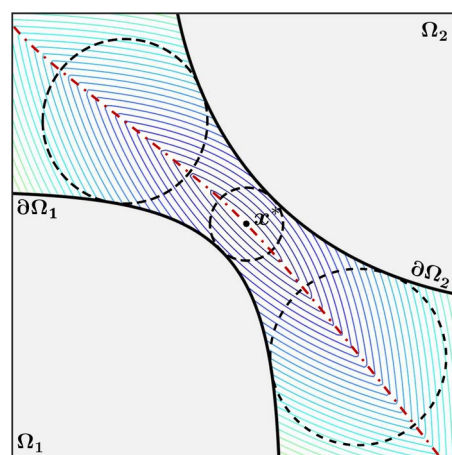


Fig. 13 Illustration of a 2D GDF ϕ_G for two arbitrary superellipses. The medial axis (dashed red line) represents the points where $\phi_i(x) = \phi_j(x)$. The minimum distance point lies on the medial axis and represents the minimum value of the resultant GDF

first-order optimality condition for minimum distance with respect to the contact points:

$$\begin{aligned} \nabla \mathcal{L}(\mathbf{p}_1, \mathbf{p}_2) &= \begin{bmatrix} \frac{\partial \mathcal{L}}{\partial \mathbf{p}_1} \\ \frac{\partial \mathcal{L}}{\partial \mathbf{p}_2} \end{bmatrix} \\ &= \begin{bmatrix} -\mathbf{r}_{12} + \nu_1 \nabla f_1(\mathbf{p}_1) \\ \mathbf{r}_{12} + \nu_2 \nabla f_2(\mathbf{p}_2) \end{bmatrix} = \mathbf{0} \end{aligned} \quad (19)$$

Combining the above conditions and evaluating them at optimal points $\{\mathbf{p}_1^c, \mathbf{p}_2^c\}$ (closest contact points) yields:

$$\begin{aligned} \mathbf{r}_{12} &= \nu_1 \nabla f_1(\mathbf{p}_1^c) = -\nu_2 \nabla f_2(\mathbf{p}_2^c) \\ &:= \nu_1 \mathbf{n}_1(\mathbf{p}_1^c) = -\nu_2 \mathbf{n}_2(\mathbf{p}_2^c) \end{aligned} \quad (20)$$

$$\begin{aligned} \therefore \mathbf{r}_{12} &\parallel \mathbf{n}_1(\mathbf{p}_1^c) \quad \text{and} \quad \mathbf{r}_{12} \parallel \mathbf{n}_2(\mathbf{p}_2^c) \\ \Rightarrow \mathbf{n}_1(\mathbf{p}_1^c) &\parallel \mathbf{n}_2(\mathbf{p}_2^c) \end{aligned} \quad (21)$$

Consider all vectors scaled to unit norm. In fact, from Eq.(21), one can effectively conclude that contact points must share a common, parallel normal for the minimum distance condition (MDC). However, additional geometrical relations arise from this derivation. These include parallelism and anti-parallelism conditions between the distance vector and the normal vectors at the contact points, as referred to in [31, 32, 64]. The full constraints for MDC are then stated as follows:

$$\begin{aligned} c_1 &: f_{SE}(\mathbf{p}_i^c, \lambda_i) - 1 = 0 \\ c_2 &: f_{SE}(\mathbf{p}_j^c, \lambda_j) - 1 = 0 \\ v_1 &: \mathbf{n}_i \cdot \mathbf{n}_j + 1 = 0 \quad \Rightarrow \quad \mathbf{n}_i \uparrow \downarrow \mathbf{n}_j \\ v_2 &: \mathbf{n}_i \cdot \mathbf{r}_{ij}^c - 1 = 0 \quad \Rightarrow \quad \mathbf{n}_i \uparrow \uparrow \mathbf{r}_{ij}^c \\ v_3 &: \mathbf{n}_j \cdot \mathbf{r}_{ij}^c + 1 = 0 \quad \Rightarrow \quad \mathbf{n}_j \uparrow \downarrow \mathbf{r}_{ij}^c \end{aligned} \quad (22)$$

where constraints c_1 and c_2 represent the boundary constraints defined by Eq.(2), ensuring primal feasibility, and v_1, v_2 and v_3 represent the strong geometric constraints written in terms of the inner product of the normal vectors and the distance vector r_{ij}^c , ensuring global minimum distance. The vector notation $\uparrow\uparrow$ represents vector parallelism while $\uparrow\downarrow$ represents anti-parallelism. All vectors are scaled to unit norm.

Houlsby [65] proposed an approach that focuses on determining a midpoint between the bodies that is closest to both rather than directly seeking the closest contact points, assuming that the involved potential functions are strictly convex, as is the case for convex superellipses. This was achieved through unconstrained optimization of the geometric potentials of the bodies involved. The proposed

approach in this research is directly inspired by this midpoint concept. Since the geometries can be modeled by their respective SDFs, which inherently capture the properties of distance metrics, we extend this method to leverage the SDF framework for improved accuracy and efficiency. Specifically, we incorporate the convexity and gradient properties of SDFs into the contact detection process, enabling robust identification of contact points and respective boundary normals. For this, the notion of Gap Distance Field (GDF) is introduced.

The GDF represents a simple, yet useful, mathematical formulation derived from the individual SDFs of the considered bodies from which it is possible to quantify the spatial relationship between them. From CSG, the GDF can be defined as the Boolean intersection of the individual SDFs, or as its equivalent pointwise maximum:

$$\begin{aligned}\phi_G(\mathbf{x}) &= \phi_i(\mathbf{x}) \cap \phi_j(\mathbf{x}) \\ &= \max \{ \phi_i(\mathbf{x}), \phi_j(\mathbf{x}) \}\end{aligned}\quad (23)$$

Here, we recall some fundamental facts about convexity that will show to be beneficial for posterior usage of the GDF:

1. SDFs of convex bodies retain their convexity [66].
2. The pointwise maximum operation of convex sets preserves the convexity of the inputs [67].

Consequently, for convex bodies, the resulting GDF as defined in Eq.(23), remains convex, ensuring a unique global minimum and eliminating the risk of local minima [67, 68]. To illustrate the abovementioned concept, Fig. 11 presents an example of two circles whose exact SDFs, given by $\phi = \|x_q - r_x, y_q - r_y\| - R$, are evaluated at the 1D level cut $y = 0$. As depicted in Fig. 11, formulating an optimization problem on $\phi_G(\mathbf{x})$ allows to identify the closest mid-point to the two surfaces by finding $\mathbf{x}^*$ that minimizes $\phi_G(\mathbf{x})$. Hence, the computation of $\mathbf{x}^*$ can be formulated as a min-max problem such as $\mathbf{x}^* = \operatorname{argmin} \max \{ \phi_i(\mathbf{x}), \phi_j(\mathbf{x}) \}$. For the present research, we call the solution $\mathbf{x}^*$ the Apparent Contact Point (ACP).

For separation cases, the minimum distance is given by $2\phi_G(\mathbf{x}^*)$, as illustrated in Fig. 12(a), while for overlap conditions, Fig. 12(b), $\phi_G(\mathbf{x}^*)$ represents the radius of maximum inscribed circle within the intersection zone, centered at $\mathbf{x}^*$, also called the Chebyshev Center [69]. Apart from computing the shortest distance between the boundaries, the sign of $\phi_G(\mathbf{x}^*)$ establishes an efficient query of whether separation, perfect contact, or overlap occurred.

However, this only measure returns a lower bound approximation of the overlapping distance: for sharp and large overlaps, this measure is always an underestimator of the minimum translational depth, but it agrees on some

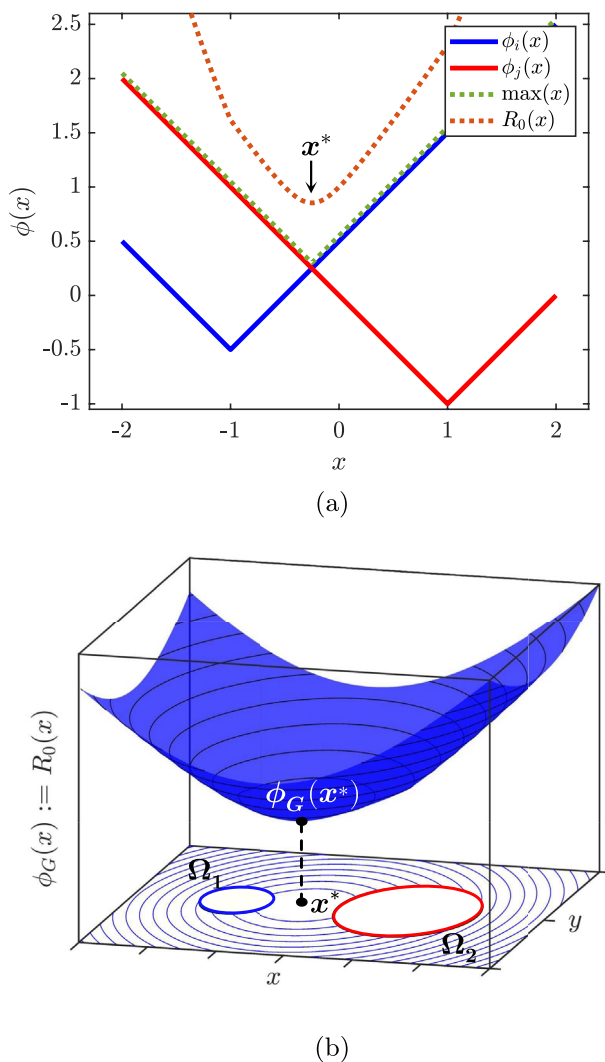


Fig. 14 **a** Illustration of the smooth R-disjunction function defined by Eq.(24), with $\alpha = 0$, compared with hard max Eq.(23), and **b** Generalization of the GDF concept for two arbitrary convex bodies. Contact point $\mathbf{x}^*$ corresponds to the solution of the minimization problem of ϕ_G

overlapping instances [18]. In fact, it can be shown that the CN condition is only fulfilled for smooth penetrations, that is if the penetration distance is small compared to the bodies' sizes and to the minimum radius of curvature of their boundaries [30].

Points that are equidistant to two convex surfaces, that is, points where $\phi_i(\mathbf{x}) = \phi_j(\mathbf{x})$, constitute the medial axis of the contact interface. The use of the hard max operator results in a C^1 -discontinuity of the GDF on its medial axis [70], as depicted in Fig. 13. This discontinuity is inherently linked to the mathematical definition of distance formulation and cannot be eliminated, even with local mesh refinement [71]. From an optimization perspective, this discontinuity can slow global convergence because of the non-differentiability of the GDF at points on the medial axis which creates challenges for gradient-based methods. On the other hand, for derivative-free optimization methods, this may lead to inefficient steps when encountering sharp changes in the function's local behavior, hampering convergence of the optimization procedure.

To overcome the shortcomings of the use of the hard max into the optimization procedure, alternative, smoother formulations must be introduced. The general concept of smooth-maximum serves as a smooth approximation of the hard maximum function while retaining its convexity. This smoothness makes it valuable for convex optimization algorithms. Among various options, such as LogSumExp or the Boltzmann operator, R-functions stand out as a particularly effective choice enabling smooth and purely algebraic Boolean operation [72]. Originally introduced by Rvachev [73], R-functions are a class of real-valued functions characterized by the property that their sign is entirely determined by the signs of their arguments. In other words, R-functions serve as Boolean switching operators, changing their sign only when the signs of their arguments change. Refer to the work of Shapiro for a deeper understanding of these functions [74, 75]. Over the years, R-functions have gained strong interest in fields such as geometric modeling [76, 77], computational physics [78], and in many other areas of computational engineering and mechanics [58, 79, 80]. One of the simplest R-functions is the R-disjunction, which is the counterpart of the max function and equivalent to Boolean union, and can be generalized as follows:

$$R_\alpha(x_1, x_2) = \frac{1}{1 + \alpha} \left(x_1 + x_2 + \sqrt{x_1^2 + x_2^2 - 2\alpha x_1 x_2} \right) \quad (24)$$

with $-1 < \alpha < 1$

The construction of such functions is based on the notion of triangle inequality. Considering x_1 and x_2 as the lengths

of sides of a triangle, the length of the third side is determined via cosine law as presented in the argument under the square-root of Eq.(24) with α being the cosine of the angle between the two sides. By setting $\alpha = 1$, the hard max function is recovered, corresponding to a singular triangle. The resulting function is continuous but is not differentiable along the line $x = y$. For the special case of $\alpha = 0$, the constructed triangle becomes a right triangle, and the resulting R_0 -function becomes analytic everywhere, except at the origin.

This can then be applied as a new representation of the gap distance field $\phi_G(\tilde{\phi}_i, \tilde{\phi}_j) := R_0(\tilde{\phi}_i, \tilde{\phi}_j)$ by substituting the R_0 -function arguments with the approximate SDFs of the two interacting bodies evaluated at a spatial point $\mathbf{x}$. The resulting field ϕ_G provides a smooth representation of the gap between the two bodies at each point, incorporating the sign-preserving property of R_0 -functions: negative values indicate penetration, zero represents contact, and positive values denote separation. This property is particularly advantageous for contact detection, as penetration queries can be efficiently performed by simply evaluating the sign of $\phi_G(\mathbf{x}^*)$. The R_0 -function then presents to be a good approximate and smooth alternative to the hard max function to generate the GDF. Figure 14 presents the comparison between the original hard max function and the smoothed maximum given by the R_0 -function for the same example presented in Fig. 11.

The apparent contact point $\mathbf{x}^*$ can be determined via numerical minimization of the constructed GDF. For this purpose, the Nelder-Mead optimization algorithm [81] was employed due to its suitability for discrete, convex functions. Although convex, the numerical representation of the present GDF is discrete, rendering gradient-based methods less suitable due to the absence of explicit gradient information and the existence of potentially noisy finite-difference approximations. The Nelder-Mead method, being a derivative-free method, efficiently converges to the global minimum by relying solely on function evaluations, which is ideal for the present, low-dimensional, optimization problem. Additionally, the convexity of the GDF ensures fast convergence of the Nelder-Mead algorithm, ensuring stable and accurate identification of the ACP. For the initial iteration, in the absence of prior information, the initial guess is chosen as the geometric bisection of the vector joining the centroids of the interacting bodies. In the context of dynamic or static nonlinear simulations, with multiple steps, under the assumption of small translations and rotations, the initial guess for time-step Δt can be initialized using the previous ACP solution $\mathbf{x}_{\Delta t-1}^*$.

5.4 Contact pair determination

Once the GDF minimization problem is solved and the apparent contact point $\mathbf{x}^*$ is obtained, contact information, such as the contact pair, contact normals, and minimum distance/overlap can be computed.

For a general arbitrary surface, determining the closest point projection of a given query point to the interface requires the use of an iterative procedure that minimizes the distance between those points, as presented in Eq.(9). The solution of this problem can be obtained numerically by, i.e., the Newton–Raphson method [82] or other optimization method [83].

Let's recall the orthogonality property of the gradient, which states that the gradient of a scalar field is always perpendicular to its isolines. Since surface normals are defined to be perpendicular to the surface, they necessarily coincide with the direction of the gradient of the scalar field. Consequently, closest contact points $\mathbf{p}_1^c$ and $\mathbf{p}_2^c$ must be located along the gradient of $\tilde{\phi}_i$ and $\tilde{\phi}_j$, respectively. Additionally, from the Eikonal property of exact SDFs, the closest contact points $\mathbf{p}_1^c$ and $\mathbf{p}_2^c$ can be obtained by performing a gradient projection operation without need of any optimization algorithms [84]. The SDF gradient projection operation is defined as follows:

$$\mathbf{p}(\mathbf{x}^*) = \mathbf{x}^* - \nabla\phi(\mathbf{x}^*)\phi(\mathbf{x}^*) \quad (25)$$

Although the stated projection is valid under the Eikonal condition, additional rescaling of the normal to unit length is needed due to numerical approximation of the gradient. The gradient $\nabla\phi(\mathbf{x}^*)$ can be numerically approximated through a 5-point stencil central finite difference scheme as follows:

$$\nabla\Phi(\mathbf{x}^*) \approx \begin{bmatrix} \frac{\Phi_{\mathbf{x}_x^*, \mathbf{x}_y^* + \Delta h} - \Phi_{\mathbf{x}_x^*, \mathbf{x}_y^* - \Delta h}}{2\Delta x} \\ \frac{\Phi_{\mathbf{x}_x^*, \mathbf{x}_y^* + \Delta h} - \Phi_{\mathbf{x}_x^*, \mathbf{x}_y^* - \Delta h}}{2\Delta y} \end{bmatrix} \quad (26)$$

where Δx and Δy denote the grid spacing along the coordinate axes. Due to numerical attenuation arising from the finite difference approximation of the gradient, the projection operation can be successively applied to ensure convergence, as defined in Eq.(27). Different boundary convergence criteria such as: zero distance to boundary $|\phi(\mathbf{p})|$ or by the implicit condition given by Eq.(2) can be considered. Fortunately, given the distance metric properties, no more than two iterations should be necessary for correct convergence [85].

$$\mathbf{p}^{i+1} \leftarrow \mathbf{p}^i - \nabla\phi(\mathbf{p}^i)\phi(\mathbf{p}^i) \quad (27)$$

The boundary normal vector of contact point can also be obtained through the properties of its SDF. In fact, at the zero-level set, the gradient $\nabla\phi(\mathbf{x})$ corresponds directly to the outward normal vector for the given boundary. For a fine enough boundary discretization, the normal vector can also be obtained (with good approximation error) from the implicit formulation of the superellipse at the point of interest. For a rotated and non-centered superellipse, the components of the outward normal, in the local FoR, are given by the following matrix form:

$$\nabla f_{SE}(x, y) = \frac{2}{\varepsilon} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \frac{1}{a} |X|^{\frac{2}{\varepsilon}-1} \operatorname{sgn}(X) \\ \frac{1}{b} |Y|^{\frac{2}{\varepsilon}-1} \operatorname{sgn}(Y) \end{bmatrix} \quad (28)$$

where

$$X = \frac{(x - \mathbf{r}_p^x) \cos\theta + (y - \mathbf{r}_p^y) \sin\theta}{a}$$

$$Y = \frac{-(x - \mathbf{r}_p^x) \sin\theta + (y - \mathbf{r}_p^y) \cos\theta}{b}$$

While parametric forms also offer an alternative route to computing boundary normals, they are highly sensitive to variations in the contact point coordinates, particularly for superellipses with low squareness coefficients $\varepsilon \ll 1$, where the sensitivity of point location w.r.t. the parametric variable is high and can amplify numerical instabilities. Hence, small errors in the computed contact points can lead to significant deviations in the normal vector direction. The full framework of the proposed algorithm can be expressed as the following pseudo-code:

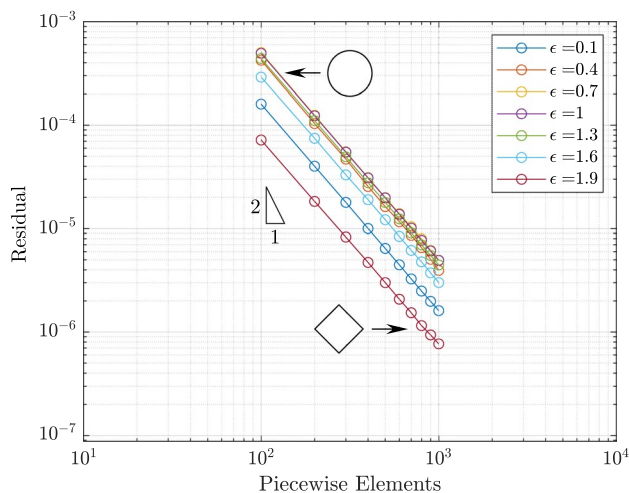


Fig. 16 Effect of squareness coefficient and discretization level on the integral L_1 -norm of the SDF error

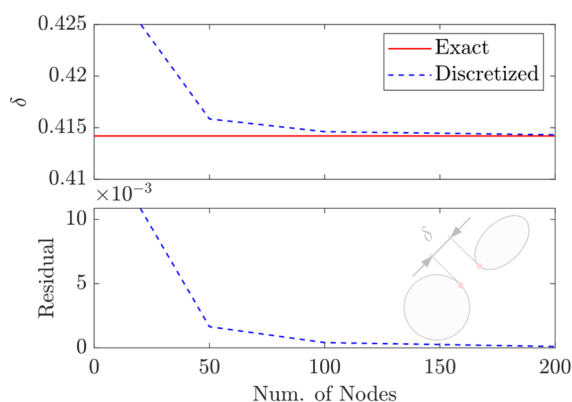


Fig. 17 Effect of discretization level on the L1-norm of the minimum distance between two superellipses defined by $\lambda_1 = \{0.5, 0.5, 1\}$ $q_1 = \{0, 0, 0\}$ and $\lambda_2 = \{0.5, 0.3, 1\}$ $q_2 = \{1, 1, \frac{\pi}{4}\}$

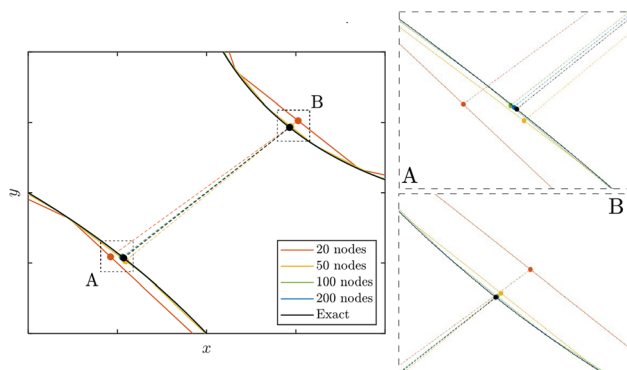


Fig. 18 Effect of discretization level in the computation of contact points and distance vector for scenario between two superellipses defined by $\lambda_1 = \{0.5, 0.5, 1\}$ $q_1 = \{0, 0, 0\}$ and $\lambda_2 = \{0.5, 0.3, 1\}$ $q_2 = \{1, 1, \frac{\pi}{4}\}$

```

1: Inputs:  $\{a_i, b_i, \varepsilon_i, \mathbf{r}_i, \theta_i\}, \{a_j, b_j, \varepsilon_j, \mathbf{r}_j, \theta_j\}$ 
2:  $\tilde{\phi}_i, \tilde{\phi}_j \triangleright$  Algorithm 1
3:  $\tilde{\phi}_G = R_0(\tilde{\phi}_i, \tilde{\phi}_j) \triangleright$  Eq.(24)
4:   or  $\tilde{\phi}_G = \max(\tilde{\phi}_i, \tilde{\phi}_j) \triangleright$  Eq.(23)
5: if  $t = 0$  then
6:    $\mathbf{x}_0 = \frac{\mathbf{r}_i + \mathbf{r}_j}{2}$ 
7: else
8:    $\mathbf{x}_0 = \mathbf{x}_{\Delta t-1}^*$ 
9: end if
10:  $\mathbf{x}^* = \min \tilde{\phi}_G \triangleright$  Nelder-Mead Algorithm
11:  $\{\mathbf{p}_1^c(\mathbf{x}^*), \mathbf{p}_2^c(\mathbf{x}^*)\} \triangleright$  Eq.(27)
12:  $\delta \leftarrow \text{sgn}(\phi_G(\mathbf{x}^*)) \cdot \|\mathbf{p}_1^c(\mathbf{x}^*) - \mathbf{p}_2^c(\mathbf{x}^*)\|$ 
13:   or  $\delta = \phi_G(\mathbf{x}^*)$ 
14:  $\{\mathbf{n}_1, \mathbf{n}_2\} \triangleright$  Eq.(28)
15: return  $\delta, \mathbf{p}_1^c(\mathbf{x}^*), \mathbf{p}_2^c(\mathbf{x}^*), \mathbf{n}_1, \mathbf{n}_2$ 

```

Fig. 15 Minimum Distance/Overlap, Contact Pair, and Contact Normals

6 Numerical experiments

6.1 Effect of discretization and GDF formulation

At first, being the proposed methodology based on the computation of approximate SDF from discretized boundaries, it is important to assess the effect of the discretization level on the obtained results. The global residual error, quantified as the integral of the L_1 -norm of the SDF error over the analysis domain, as defined in Eq.(29), was analyzed for various superellipse geometries under increasing boundary discretization levels. Unlike point-based metrics (e.g., maximum or relative error), this residual error measures the total deviation of the computed SDF from the “ground truth” SDF over the entire domain. This provides a more comprehensive picture of how well the SDF performs across the entire geometry, rather than just at isolated points. This measure averages the deviations across the domain, reducing the impact of outliers that might overly influence maximum or pointwise error metrics.

$$\int_{\Omega} |\tilde{\phi}_i(x, y) - \phi_i(x, y)| dx dy \quad (29)$$

where $\tilde{\phi}_i(x, y)$ is the approximated SDF of the discretized boundary and $\phi_i(x, y)$ represents a “ground truth” SDF of the corresponding discretized boundary with 10 000 nodes. Both are computed using Algorithm 1. Figure 16 illustrates the residual error for different shape coefficients ε w.r.t the number of nodes. All cases used unit semi-axes and were axis-aligned and centered at the origin.

Results reveal that integral residual error decreases with increasing discretization for all geometries, as expected. Furthermore, the SDF residual errors exhibit a second-order rate of convergence, within the range of exponents ε for convex bodies. Geometries with extreme exponents ($\varepsilon = 0.1$ and $\varepsilon = 1.9$) exhibit lower residual errors compared to intermediate cases ($\varepsilon = 0.4, 0.7, 1.3, 1.6$) and even the circular geometry ($\varepsilon = 1.0$). This behavior can be attributed to the discretization method that employed the parametric formulation of the superellipse, which inherently concentrates nodes in higher curvature regions. For shapes with sharp corners or zones of high curvature, such as the rectangle-like geometry ($\varepsilon = 0.1$) and the rhombus-like geometry ($\varepsilon = 1.9$), this adaptive discretization ensures more accurate SDF approximation in those challenging regions. For the circular geometry ($\varepsilon = 1.0$), the curvature is uniform along the boundary, requiring consistent resolution everywhere. This uniformity results in higher residual errors compared to the extreme geometries that benefit from adaptive discretization. Intermediate geometries ($\varepsilon = 0.4, 0.7, 1.3, 1.6$) combine challenges from both

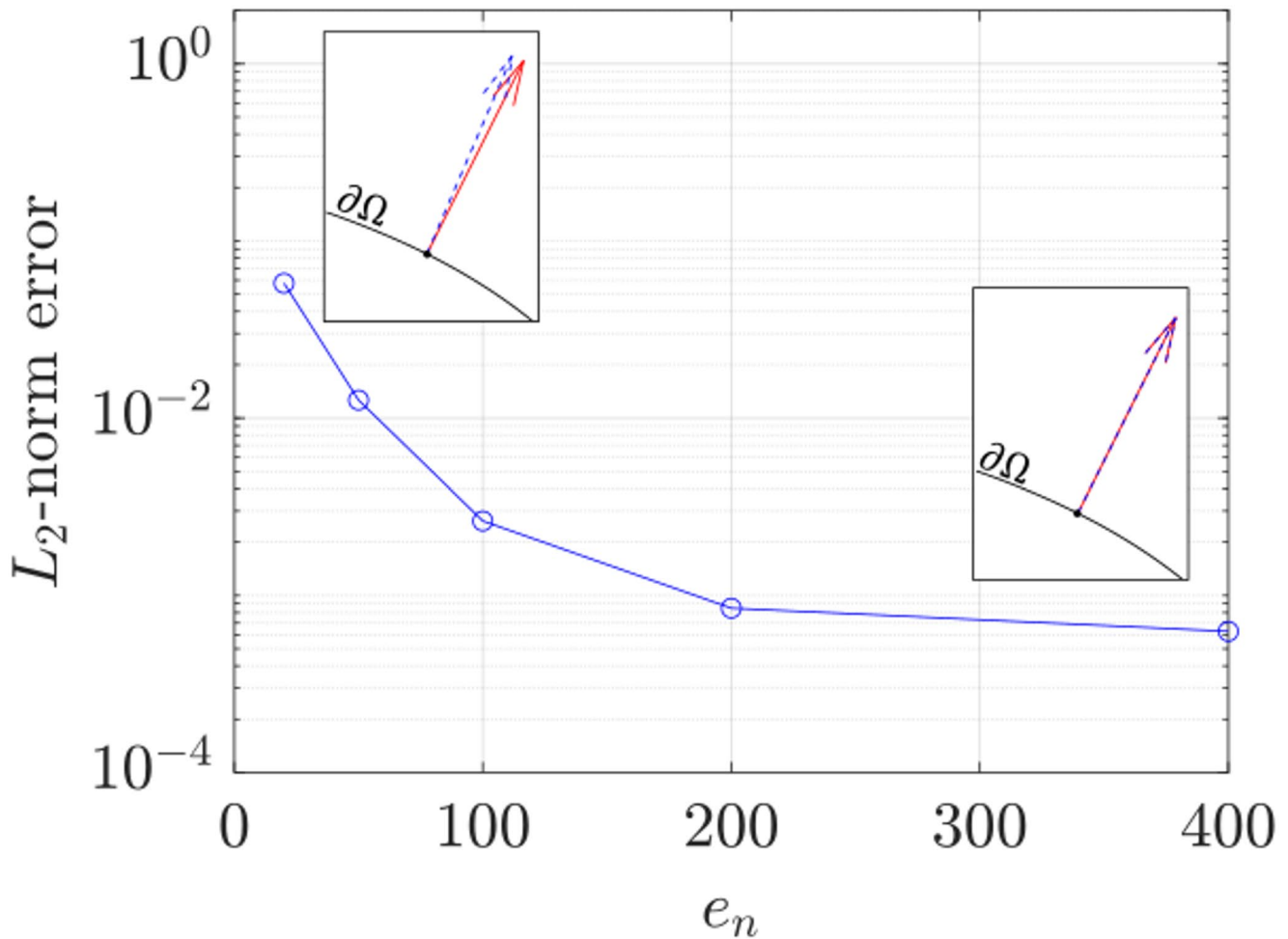


Fig. 19 Effect of discretization level on the L_2 -error norm between numerical (dashed blue line) and exact (full red line) outward normal vector of a superellipse defined by $\lambda_1 = \{0.5, 0.5, .2\}$ $q_1 = \{0, 0, 0\}$ at “corner” point

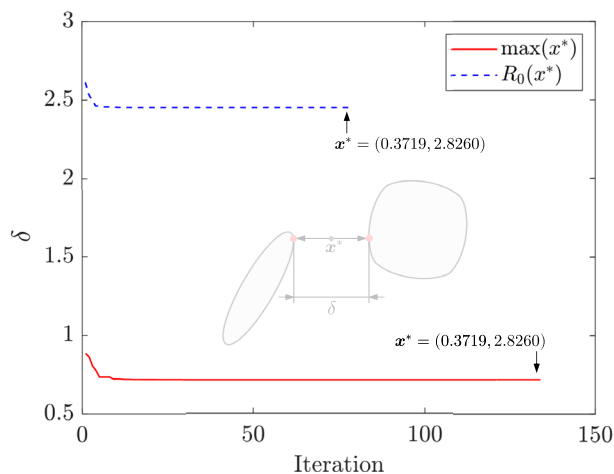


Fig. 20 Convergence comparison of the apparent contact point computation when hard max (red full line) and R_0 function (dashed blue line) are used as GDF formulation

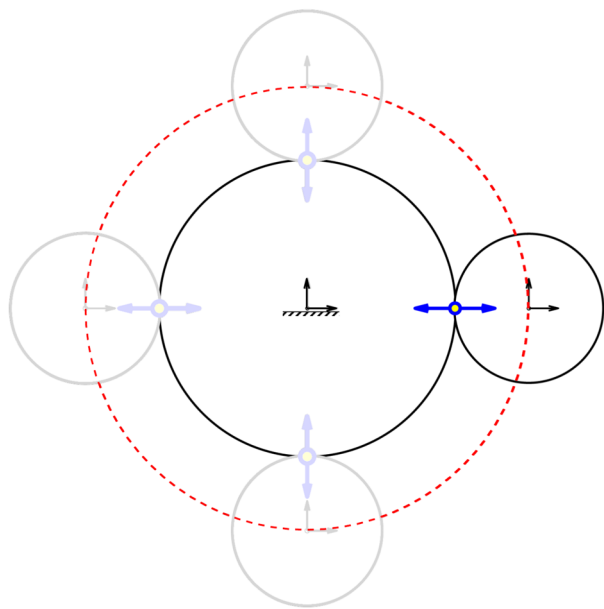
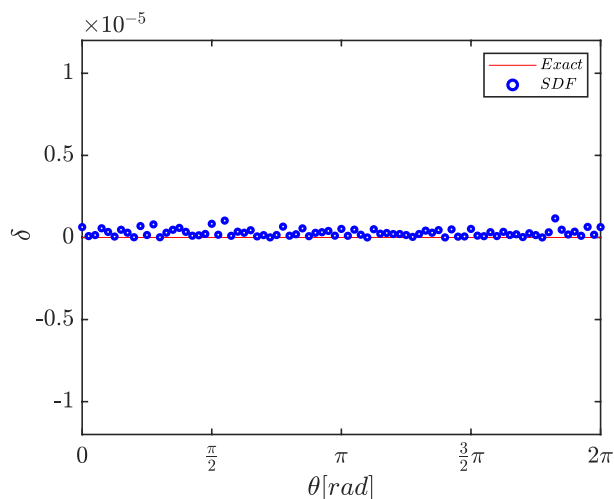
sharp corners and curvature, leading to intermediate, but still acceptable error levels.

For some specific superellipse pair configurations, an exact minimum distance can be intuitively determined, i.e. when minimum distance points coincide with the superellipse’ extremum or “corner” points. Given that, a static contact pair was considered to assess the effect of boundary discretization on the minimum distance computation. Figure 17 shows the convergence of minimum distance w.r.t to the discretization level as well as the respective residual, given by the L_1 -norm error.

For the same superellipse pair, influence of discretization on the closest contact points determination was analyzed. Results are presented in Fig. 18 and show the convergence to exact solution of computed closest contact points, and respective distance vectors, for different boundary discretization levels.

Table 1 Average results for 1000 random circle–circle cases

Case	Relative distance	Relative overlap	Normalized contact point	Avg. CPU time per contact detection
Circle–circle [27]	1.50×10^{-5}	1.75×10^{-5}	6.23×10^{-5}	68 ms

**Fig. 21** Circle–circle exact tangent contact between fixed and moving body**Fig. 22** Circle–circle exact tangent contact distance as a function of angular location

As higher discretization levels are applied, the computed minimum distance points converge toward the exact solution, as expected. Coarser discretization exhibits noticeable deviations of the distance vector, while finer ones closely align with the exact normal. Beyond 100 nodes, the improvements become marginal, as also shown in Fig. 19,

indicating that moderate discretization offers a computationally efficient balance between accuracy and cost. Additionally, the effect of nodal accuracy is noticeable in the present example, especially in superellipse Ω_2 . Indeed, when a boundary node coincides or is in the vicinity of the exact contact point, convergence accelerates, allowing good accuracy with fewer nodes. Hence, additional refinement strategies could mitigate any residual errors in cases of misaligned discretization nodes, further contributing to cost-accuracy balance.

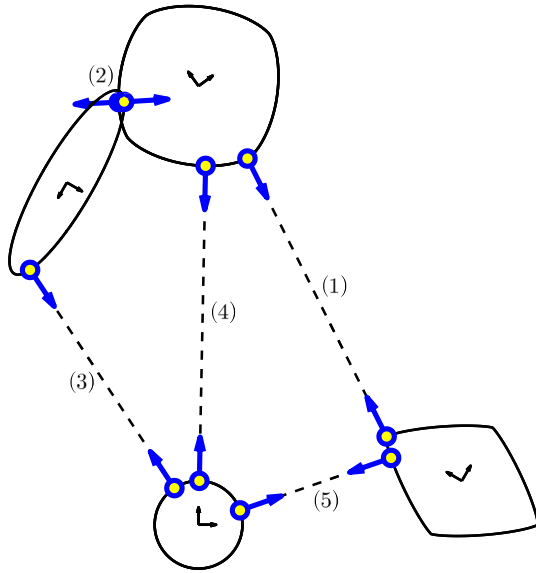
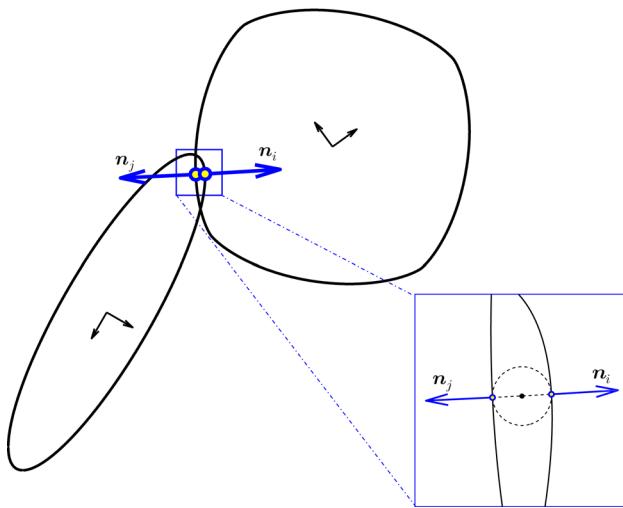
Additional evaluation of the proposed methods was done on the accuracy of the normal boundary vector calculation. As previously stated, at boundary points, the outward normal coincides with the gradient of the bodies SDF. Thus, evaluating the accuracy of the outward normal forcefully corresponds to evaluating the accuracy of the SDF gradient at boundary points. We then evaluate the gradient at “corner” point of a high-squaredness standard superellipse defined by $\lambda_1 = \{0.5, 0.5, .2\}$. This point corresponds to the region of maximum curvature on the superellipse, making it more challenging for accurate gradient computation. The gradient $\nabla\phi$ was numerically approximated by Eq.(26). The deviation in the normal vector was computed through the L_2 -norm error between the exact normal vector of the superellipse, given by Eq.(28), and the numerical normal vector. The semilog plot of Fig. 19 depicts the L_2 -norm error derived from the gradient of the SDF as a function of the number of elements in the discretized boundary.

The decreasing trend of the error curve demonstrates that as the boundary discretization becomes finer, the numerical approximation of the normal vector approaches the exact solution. The rapid decay in error indicates that the SDF-based gradient computation method achieves high alignment of the outward normal with the exact solution without requiring excessively fine meshes before plateauing, which indicates diminished returns for further refinement.

Apart from discretization, the choice of the GDF formulation can also deeply impact the convergence to correct apparent contact point. For that reason, a numerical test was performed to evaluate the effect of use of the hard max or R_0 -function as formulations for the GDF. The test performed involved bodies defined by $\lambda_1 = \{1.2, 0.3, 1\}$, $q_1 = \{0, 2, 4\pi/3\}$ and $\lambda_2 = \{1, 1, 1.3\}$, $q_2 = \{2.3, 3, \pi/5\}$, as shown in Fig. 20. A convergence tolerance of 1×10^{-12} was employed for both GDF formulation tests. Results showed that, as planned, the smoother max version of the GDF given by the R_0 -function converged more rapidly than the hard max counterpart (78 iterations against 133). Furthermore, both methods converged to the same minimum point (apparent contact point) which is a good indicator of the accuracy of

Table 2 Shape and spatial parameters for superellipse to superellipse contact

Case	$\lambda_i ; q_i$	$\lambda_j ; q_j$
1	(1, 1, 1.3); (2, 3, $\pi/5$)	(0.7, 1, 1.7); (5, $-\pi/3$)
2	(1, 1, 1.3); (2, 3, $\pi/5$)	(1.2, 0.3, 1); (0.5, 1.9, $4\pi/3$)
3	(0.5, 0.5, 1.3); (2, $-2, 0$)	(1.2, 0.3, 1); (0.8, 1.9, $4\pi/3$)
4	(1, 1, 1.3); (2, 3, $\pi/5$)	(0.5, 0.5, 1.5); (2, $-2, 0$)
5	(0.7, 1, 1.7); (4, 0, $\pi/3$)	(0.5, 0.5, 1.5); (2, $-2, 0$)

**Fig. 23** Illustration of contact points (yellow dots), their corresponding normal directions (blue arrows), and CN (dashed line) between different superellipses**Fig. 24** Detailed view of contact zone for case 2 with illustration of contact points (yellow dots), their corresponding normal directions (blue arrows), and CN (dashed line) between different superellipses

the proposed R_0 -function as alternative to the hard max function.

6.2 Validation tests

Following the evaluation of the impact of discretization on the accuracy of the proposed method, a comprehensive set of tests was conducted to assess the method's performance across various configurations of superelliptical body pairs. For consistency, a baseline discretization of 400 boundary nodes per body was adopted throughout all test scenarios. This discretization level was chosen to balance computational efficiency with sufficient resolution for accurate contact detection and geometric representation. The selection remained conservative, providing a higher resolution than strictly necessary to ensure robust and consistent results, as demonstrated in the previous section.

In the absence of a rigorous interpretation of “correct” contact points and lack of general analytical solutions, the validation of proposed algorithms for geometries other than circles or spheres, or in very specific spatial configurations of superellipsoids, becomes a hard task. As an alternative validation method, Lin and Ng [27] stated that the defined contact points and particle overlaps with different algorithms should be identical to the analytical ones in the case of two equal-sized spheres. Table 1 summarizes the average results of 1000 randomly generated circle-circle cases, highlighting the accuracy of distance computation/overlap, contact points determination, and CPU time. All numerical experiments were conducted on a laptop machine equipped with an Intel Core i9-12900 H processor with 64GB of RAM and ran in MATLAB environment.

The normalized contact point discrepancy is the error metric of the distance between the contact points determined by the proposed numerical algorithm and the analytical solution, normalized by the sphere's radius. Additionally, the relative error of both distances and overlap is defined as $(s - s_0)/s_0$, where s are computed values and s_0 are the exact solutions derived from the analytical formulas, as in [27].

To further validate the accuracy of the proposed approach, a numerical test was conducted with, still, two circular bodies, one of radius 2 and the other with radius 1. The first particle is kept fixed at global origin while the second moves along a circular path of radius $a_1 + a_2$ ensuring exact tangency of the bodies at contact point, with no rotation, as depicted in Fig. 21. Computed contact points and respective normals are also depicted.

Table 3 Results of the different superellipse to superellipse contact cases including minimum distance conditions residuals

Case	$(x_i^c); (x_j^c)$	Gap	Contact status	Elapsed time [ms]	MDC [-]
1	$(2.556, 2.173)$ $(4.143, -0.995)$	3.543	Separation	61	$c_1: 1.30 \times 10^{-12}$ $c_2: 1.20 \times 10^{-12}$ $v_1: 1.52 \times 10^{-6}$ $v_2: -5.76 \times 10^{-6}$ $v_3: 1.32 \times 10^{-5}$
2	$(1.090, 2.812)$ $(1.153, 2.815)$	-0.062	Overlap	55	$c_1: -3.95 \times 10^{-5}$ $c_2: -6.17 \times 10^{-5}$ $v_1: 1.28 \times 10^{-7}$ $v_2: -2.00 \times 10^0$ $v_3: 2.00 \times 10^0$
3	$(1.725, -1.582)$ $(0.081, 0.904)$	2.981	Separation	51	$c_1: 5.41 \times 10^{-13}$ $c_2: -5.54 \times 10^{-5}$ $v_1: 1.05 \times 10^{-5}$ $v_2: -5.33 \times 10^{-6}$ $v_3: 8.81 \times 10^{-7}$
4	$(2.077, 2.090)$ $(2.008, -1.500)$	3.591	Separation	45	$c_1: -7.03 \times 10^{-6}$ $c_2: -6.06 \times 10^{-13}$ $v_1: 2.68 \times 10^{-5}$ $v_2: -6.65 \times 10^{-6}$ $v_3: 6.73 \times 10^{-6}$
5	$(4.193, -1.236)$ $(2.473, -1.838)$	1.822	Separation	46	$c_1: -1.19 \times 10^{-5}$ $c_2: -2.73 \times 10^{-12}$ $v_1: 8.74 \times 10^{-6}$ $v_2: -2.99 \times 10^{-6}$ $v_3: 2.20 \times 10^{-5}$

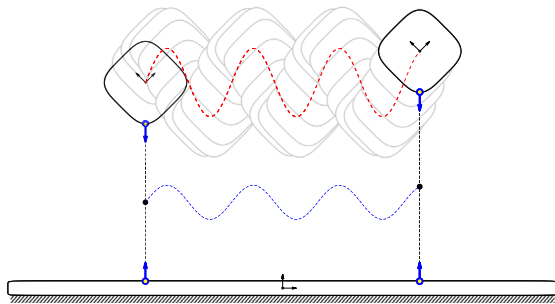
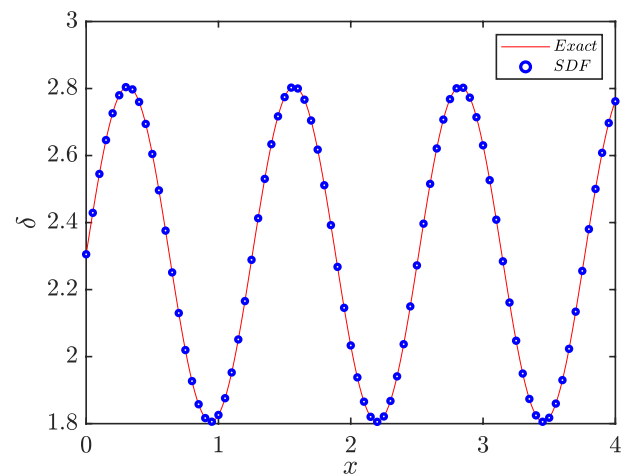
**Fig. 25** Moving contact between fixed and moving body with sinusoidal path defined by $\lambda_1 = \{4, 0.1, 0.1\}$ and $q_1 = \{3, 0, 0\}$; $\lambda_2 = \{0.5, 0.5, 0.5\}$ and $q_2^0 = \{1, 3, \pi/4\}$, respectively. Dashed red line represents the imposed motion and dashed blue line the motion of the apparent contact point

Figure 22 presents the computed signed distance, across the angular positions of the circular path. Analytically, the gap distance between the particles should be zero. Predicted results present an error on the order of 1×10^{-6} . This precision highlights the robustness of the method for accurately

**Fig. 26** Minimum distance between fixed and moving body with sinusoidal path defined by $\lambda_1 = \{4, 0.1, 0.1\}$ and $q_1 = \{3, 0, 0\}$; $\lambda_2 = \{0.5, 0.5, 0.5\}$ and $q_2^0 = \{1, 3, \pi/4\}$, respectively. Full red line represents exact minimum distance while blue points represent computed values

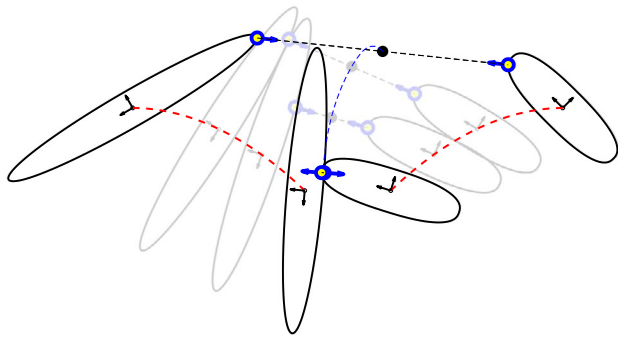


Fig. 27 Contact between two moving bodies defined by $\lambda_1 = \{0.25, 2, 1\}$ $q_1^0 = \{-3, 2, 2\pi/3\}$ and $\lambda_2 = \{0.3, 1, 0.8\}$ $q_2^0 = \{3, 2, \pi/4\}$

detecting contacts without introducing significant numerical artifacts. Further tests revealed that this error magnitude is agnostic to the scale of the particles, persisting at the same order of magnitude irrespective of the particle relative radii. This result emphasizes the scalability and reliability of the SDF approach for handling contact interactions involving particles of varying sizes.

Following the preliminary circle-circle test case, further configurations were explored to assess the robustness and versatility of the proposed contact detection methodology. For this, extended scenarios to address more complex geometries and spatial arrangements were defined, challenging the computational framework and evaluating its adaptability to diverse contact conditions. The different tested scenarios are summarized in Table 2 and depicted in Fig. 23. Table 3 summarizes the results for the tested cases.

Analyzing results for minimum distance constraints in Table 3, for almost all cases the minimum distance residuals present low values, indicating good convergence of the contact pair as well as the contact vectors. However, Case 2, which differs from the other cases due to the presence of overlap between the two superellipses, shows high residual of the minimum distance constraints. A detailed view of Case 2 is illustrated in Fig. 24 highlighting the intersecting regions. Maximum overlap points and respective contact normal are depicted, and show good alignment with the CN condition. High residuals for v_2 and v_3 constraints are due to a switching of the vector parallelism and anti-parallelism condition with the distance vector due to overlap. In fact, if only strict parallelism is considered, ignoring directionality, residuals for these constraints are 1.75×10^{-7} and -6.02×10^{-7} , showing perfect alignment of the vectors involved.

To extend the validation of the approach, a dynamic numerical test was performed to investigate the method's capability in capturing contact interactions within a time-dependent framework. In this scenario, the contact between a sinusoidally oscillating superelliptical body and a fully constrained ground body is analyzed. The parameters for

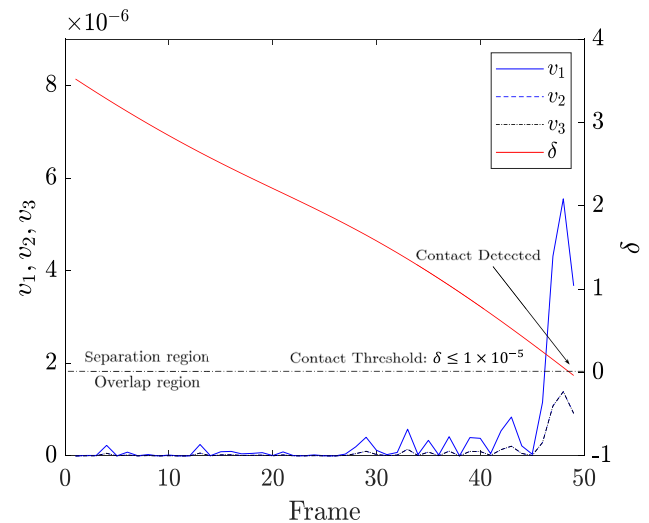


Fig. 28 Gap values (full red line) and strong geometrical constraints of minimum distance: normal vector anti-parallelism (full blue line) and normal vector to distance vector “relaxed-parallelism condition” (dashed lines)

the ground body are $\lambda_1 = \{4, 0.1, 0.1\}$ and $q_1 = \{3, 0, 0\}$; while the moving body is defined as $\lambda_2 = \{0.5, 0.5, 0.5\}$ and $q_2^0 = \{1, 3, \pi/4\}$. The red dashed line represents the imposed motion of the particle's centroid, while the blue dashed lines denote the dynamic location of the apparent contact point determined by the proposed method.

For clarity, contact pairs and respective normals are depicted only at the initial and final time steps.

Building upon the previous validation, a complementary test was performed to assess the method under more dynamic conditions, introducing initial linear velocities, constant angular velocities, and gravitational effects. The bodies are defined as $\lambda_1 = \{0.25, 2, 1\}$, $q_1^0 = \{-3, 2, 2\pi/3\}$ and $\lambda_2 = \{0.3, 1, 0.8\}$, $q_2^0 = \{3, 2, \pi/4\}$ and the $\dot{x}_1^0 = \{5, 0\}$ m/s and $\dot{x}_2^0 = \{-5, 0\}$ m/s and $\omega_1 = 2$ rad/s and $\omega_2 = 1$ rad/s. The simulation was carried out using a fixed time step of 0.01 s. A contact threshold was set to 1×10^{-5} , triggering a contact flag when the computed distance was less than or equal to the specified limit. Figure 27 illustrates the displacement of the moving bodies over some frames, along with the initial and final contact pairs and their corresponding outward normal.

Figure 28 presents the numerical results for gap distance computed at each frame/time-step until the contact flag was triggered. Additionally, it illustrates the residuals associated with the strong geometric constraints for the minimum distance problem, as formulated in Eq.(11). The plot in Fig. 28 shows that the gap values steadily decrease over time until contact is detected around frame 48 when the gap reaches the contact threshold, triggering the stop of the simulation.

The results highlight effective constraint satisfaction, with consistent low values of the low geometric constraints

residuals, and precise gap evaluation throughout the simulation, confirming the method's capability to accurately capture the contact instant and compute relevant contact quantities accurately. **Note:** *In overlap condition, the residuals of strong geometric constraints were relaxed to consider "strict parallelism" to enhance the clarity of the plot. This adjustment was made without compromising the accuracy or generality of the results.*

7 Discussion and conclusions

A new optimization-based methodology for computing the minimum distance between convex superelliptical bodies was proposed leveraged by the introduction of the notion of Gap Distance Field. The numerical experiments demonstrated excellent agreement with theoretical predictions, underscoring the accuracy and reliability of the proposed methodology.

Across various test cases, the method consistently identified contact points and calculated minimum distances, presenting average errors with 10^{-6} order of magnitude for the geometrical minimum distance constraints. This robustness highlights the method's potential to handle a wide range of applications in computational contact mechanics. Moreover, the proposed approach demonstrated the ability to maintain numerical stability and convergence, even under challenging conditions such as near-tangential contacts or cases involving nearly coincident boundaries and also overlap cases.

The versatility of the methodology was also demonstrated through its ability to adapt to different superellipsoidal geometries, maintaining accuracy regardless of the parameterization of the convex shapes. As expected, the performance of the proposed methodology relies heavily on the boundary discretization level, as finer discretization results in more accurate representations of the geometry and resultant distance fields. However, the general methodology is versatile and can be integrated with any algorithm capable of generating SDFs of the bodies, such as analytic distance functions, voxels, neural network representations, or others, provided that the computed fields retain the fundamental properties of Euclidean distance fields, such as continuity and monotonic growth with distance from the boundary.

Although the methodology has been validated only for 2D scenarios, its scalability to 3D contact problems involving superellipsoids is straightforward, as the convexity properties of the bodies ensure the consistency of the contact detection process in higher dimensions. The authors acknowledge the potential for further improvements in the implemented algorithms to enhance computational efficiency. Future work will focus on accelerating the contact

detection process through the development of optimized algorithms for SDF and contact point computations, leveraging strategies such as adaptive resolution techniques and machine learning approaches. In addition, integration of the proposed contact detection framework into a fully dynamic simulation environment where contact forces and system response are coupled will be pursued. This will allow for a detailed assessment of the method's accuracy and stability in simulation environments, with particular consideration of its robustness and scalability.

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